

CS 285 HW01 - Imitation Learning Report

MSE and Flow-Matching Action-Chunking Policies on Push-T

1. Training Results

Both policies learned the Push-T task successfully. The **MSE policy** improved quickly early in training, reaching about **0.59 reward at 20K steps** and finishing at about **0.69**. The **flow-matching policy** improved more steadily, reaching about **0.54 at 30K**, **0.71 at 40K**, and **0.80 at 70K steps**. Thus, flow matching achieved the better final evaluation performance.

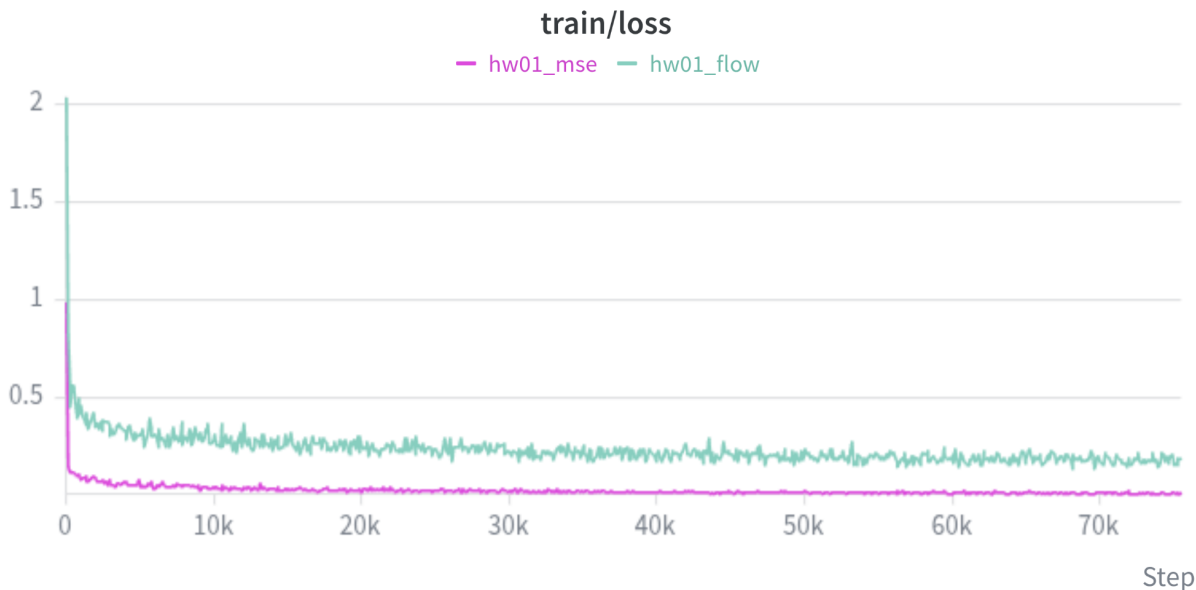


Figure 1. Training loss for MSE and flow matching. The loss values are not directly comparable because the policies optimize different objectives.

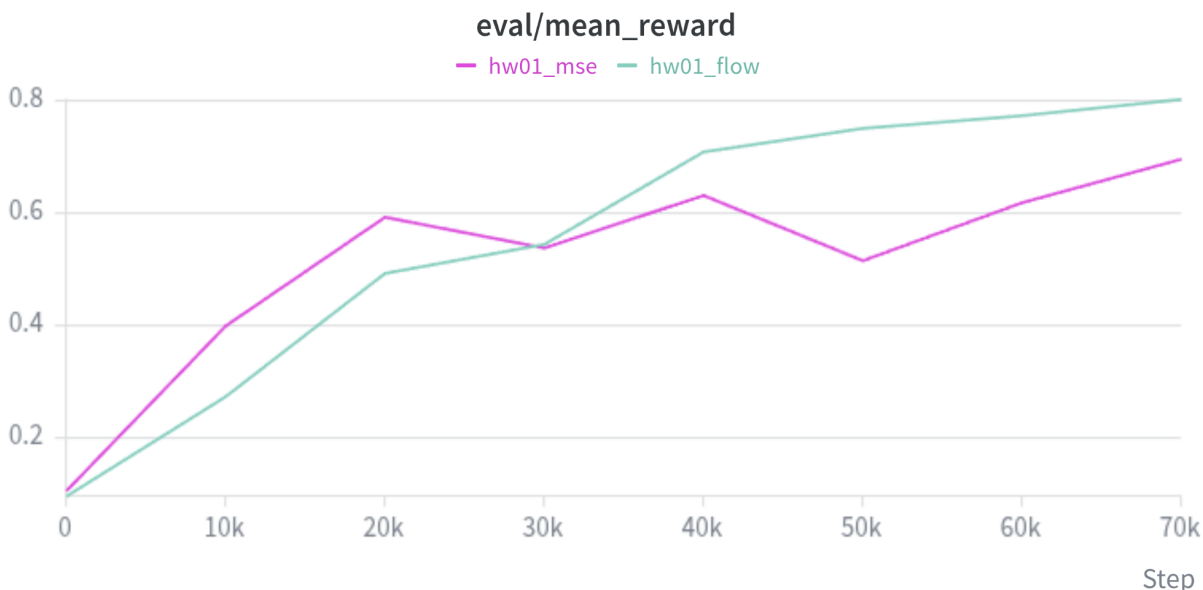


Figure 2. Evaluation reward comparison. MSE improves faster early, while flow matching reaches the higher final reward.

2. Policy Architecture

Both policies use a small MLP with **3 fully connected hidden layers**, **256 units per layer**, **ReLU** activations, a linear output layer, and **chunk size 8**. The MSE policy directly predicts the action chunk and is trained with Adam and an MSE objective. The flow-matching policy additionally takes the noisy action chunk and flow timestep τ as inputs and predicts the velocity used to transform noise into an action chunk.

3. Comparison and Summary

The MSE policy satisfies the assignment target of reward ≥ 0.5 , while the flow-matching policy satisfies its target of reward ≥ 0.7 . MSE showed faster initial learning but more fluctuation later. Flow matching improved more consistently and achieved the best final reward (**0.80 vs. 0.69**). The lower numerical MSE training loss does not mean that MSE performs better, because the two methods use different loss functions; evaluation reward is the meaningful performance comparison.

Policy	Final reward	Main observation
MSE	~0.69	Fast early learning; some later fluctuation
Flow Matching	~0.80	Steadier improvement; better final performance